

Lunchbox Build Details

I reused the **four pedals** from the prior pedal controller project, which were still relatively new. Each pedal ends in a male monophonic connector.

To plug my 4 pedals into, I used **4 monophonic headphone jacks**.

I used a **large enclosure**, measured across the back of it for the monophonic headphone jack holes, and drilled holes using a **center punch**, a **small drill bit**, then a **step-up drill bit** after using a **caliper ruler** to measure the width of the headphone jacks. For the USB cable, I used a **snap-in strain relief bushing**. For the USB cable hole, I used the same step-up drill bit after using the caliper ruler to measure the width of the cable + the bushing, then I used the same step-up drill bit to drill the hole for the USB cable and the bushing. The hole was a little loose, so to keep the bushing in place, I used a **small zip-tie** on the inside of the hole.

I used **4 stand-offs** as a platform for the **Arduino Leonardo**, after drilling small holes to accommodate them.

To replace the breadboard, I used a **screw terminal shield** which sits on top of the Leonardo.

From the prior project, I have a bunch of **jumper wires** left over from buying way too much stuff that time. I screwed **four yellow long M-F jumper wires** into the screw terminal shield's "A0-A3" inputs, and I screwed **four black M-F jumper wires** into the 3 screw terminal shield's "GND" connectors, jamming two of them together until they both fit.

Before installing the headphone jacks, I used a **soldering iron** and **some rosin-core solder**, both of which were left over from the prior project, to solder **four short yellow staple-shaped jumper wires** into each of the headphone jacks' signal lugs, and **four short dark blue staple-shaped jumper wires** onto the "GND" lugs.

I connected the female ends of each of the "A0-A3" screwed-in signal wires to its corresponding lug-soldered "signal" jumper wires. I did the same for the "GND" screwed-in wires, and the lug-soldered "GND" jumper wires.

It was then time to connect my **MIDI cable** from the Leonardo to my development PC. I uploaded the **signal test sketch** to the Leonardo, and opened the serial monitor to confirm the pedals were all behaving properly when pressed (0) and when not pressed (1).

I switched to the **MIDI driver sketch** to the Leonardo, and plugged my MIDI cable from the Leonardo into my PianoTeq PC.